Causal sufficient dimension reduction for environmental mixtures



GECC Methods Meeting

Thomas Hsiao, Ph.D.
Postdoctoral Fellow
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Emory University
Rollins School of Public Health
Department of Biostatistics and Bioinformatics

Environmental mixtures analysis

How can we quantify the causal effect of complex environmental exposures?

- Air pollution mixtures ($PM_{2.5}$, NO_2, O_3, ...)
- PFAS compounds (PFOA, PFOS, PFNA, ...)
- Metals (Pb, Cd, As, Hg, ...)

All involve multiple, highly correlated, continuous exposures



EDITORIAL • Environ Health Perspect. 2013 Jan 1;121(1):a6–a8. doi: 10.1289/ehp.1206182

Unraveling the Health Effects of Environmental Mixtures: An NIEHS Priority

Danielle J Carlin [1], Cynthia V Rider [1], Rick Woychik [1], Linda S Birnbaum [1]

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Traditionally, toxicological studies and human health risk assessments have focused primarily on single chemicals. However, people are exposed to a myriad of chemical and nonchemical stressors every day and throughout their lifetime. Some recent events that highlight the need to understand these complex exposures and their role in the etiology of disease include the *Deepwater Horizon* oil spill, the earthquake in Japan and subsequent Fukushima nuclear disaster, and the unknown environmental and health effects of hydraulic

Standard setup of the observed data

Observed data: n-vectors $O_1, ..., O_n$ sampled i.i.d. where

$$O = (Y, X, C) \sim P \in \mathcal{M}$$

- $Y \in \mathbb{R}$: outcome (binary, count, or continuous)
- $X \in \mathbb{R}^p$: continuous exposure mixture
- $C \in \mathbb{R}^q$: measured baseline covariates

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Question: What is the value of Y had we set $X = x$ for everyone in the population?

How are we currently answering this question?

Let Y^x be the potential outcome of Y after setting $X = x$.

We are interested in estimating $\mathbb{E}(Y^x)$, the **causal average dose-response**.

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Problem: Each (Bobb et al. 2015; Gibson et al. 2022; Keil et al. 2020; Rudolph et al. 2026) has their pros and cons, but none of these are estimating $\mathbb{E}(Y^x)$.

Causal inference for continuous exposures

Flexible, nonparametric, doubly robust estimators of $\mu(x) := \mathbb{E}(Y^x)$ are available.

Let $m(x, c) := \mathbb{E}(Y \mid X = x, C = c)$ and $\pi(x \mid c) := f_{X|C}(x \mid c)$.

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A kernel-based doubly robust estimator (Colangelo and Lee 2026) is

$$\hat{\mu}_{DR}(x) = \frac{1}{n} \sum_{i=1}^{n} \left\{ \frac{K_h(X_i - x)}{\hat{\pi}(X_i \mid C_i)} [Y_i - \widehat{m}(X_i, C_i)] + \widehat{m}(x, C_i) \right\},$$

where $K_h(u) = h^{-p} K(u/h)$, and $\widehat{m}$ and $\hat{\pi}$ are estimated flexibly using machine learning (e.g., neural network, random forest, or ensembles).

What can go wrong with multiple exposures?

The doubly robust estimator only works well if we do a good job on $m(x, c)$ and $\pi(x \mid c)$.

$$\sqrt{nh^p} \cdot \|\widehat{m} - m\| \cdot \|\hat{\pi} - \pi\| \xrightarrow{p} 0 \implies \sqrt{nh^p}\,\{\hat{\mu}_{\mathrm{DR}}(x) - \mu(x)\} \rightsquigarrow \mathcal{N}(0, V_x),$$

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Solution: Consider estimating the average dose-response in a lower dimensional space.

$$\boxed{\mathbb{E}(Y^x) = g(\beta^\top x), \qquad \beta \in \mathbb{R}^{p \times d}}$$

Sufficient dimension reduction (SDR)

There exists a rich literature on estimating β that preserves the *regression* relationship (Cook 2026):

$$\mathbb{E}(Y \mid X) = g(\beta^\top X).$$

Since β is not unique, the statistical estimand is the **central mean subspace**, $\mathcal{S}_{\mathbb{E}(Y|X)}$, the smallest subspace satisfying this identity.

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Problem: Because $\mathbb{E}(Y^x) \neq \mathbb{E}(Y \mid X = x)$ due to confounding, we cannot directly use regression SDR for our causal dose-response.

Ideal workflow of causal sufficient dimension reduction

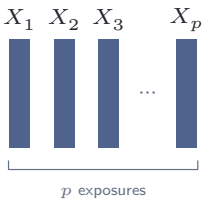
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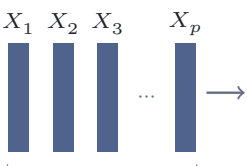


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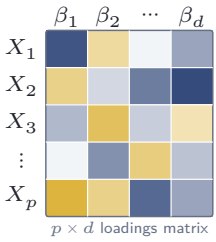
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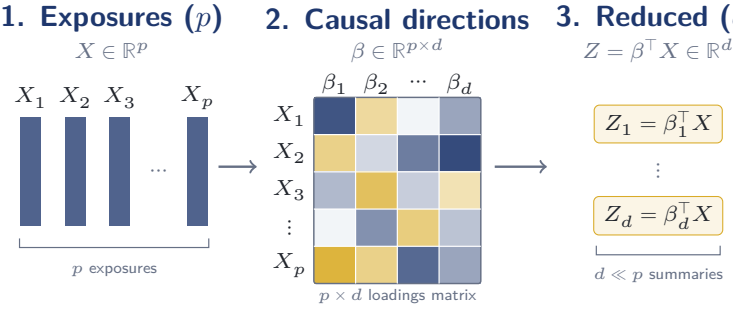
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$\beta \in \mathbb{R}^{p \times d}$



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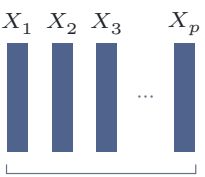


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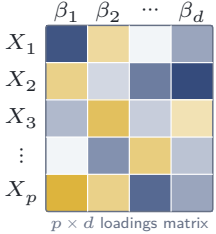
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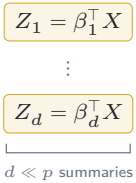
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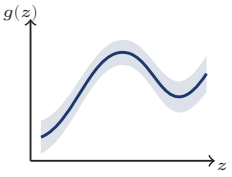
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4. Flexible response

$\mathbb{E}(Y^x) = g(z)$

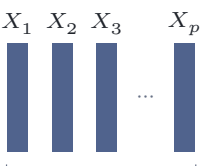


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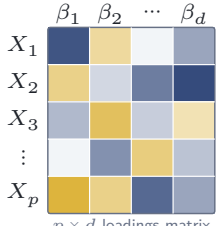
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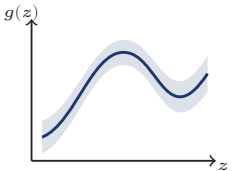
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$$Z_1 = \beta_1^\top X$$
$$\vdots$$
$$Z_d = \beta_d^\top X$$

$d \ll p$ summaries

4. Flexible response

$\mathbb{E}(Y^x) = g(z)$



Question: How can we estimate $\mathcal{S}_{\mathbb{E}(Y^x)}$ when $\mathbb{E}(Y^x) = g(\beta^\top x)$?

Estimation and Inference

A simpler approach: recast the dose-response as a regression

Suppose we can construct a variable $\widetilde{Y}$ such that

$$\mathbb{E}(\widetilde{Y} \mid X = x) = \mathbb{E}(Y^x), \qquad \forall x \in \mathcal{X}.$$

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Solution: Just run standard SDR for regression on $(\widetilde{Y}, X)$ to estimate the causal $\mathcal{S}_{\mathbb{E}(Y^x)}$!

Estimation of $\widetilde{Y}$

1. Causal dose-response:
$$\widetilde{Y} = \mu(X).$$
2. Doubly robust pseudo-outcome:
$$\widetilde{Y} = \xi = \frac{\pi(X)}{\pi(X \mid C)}\{Y - m(X, C)\} + \mathbb{E}_C[m(X, C)].$$

Both require estimation of the same nuisances

- $m(x, c) = \mathbb{E}(Y \mid X = x, C = c)$
- $\pi(x \mid c) = f_{X|C}(x \mid c)$.

Estimation of $\mathcal{S}_{\mathbb{E}(Y|X)}$ with MAVE

The minimum average variance estimator (MAVE) from Xia et al. (2002) estimates $\mathcal{S}_{\mathbb{E}(Y|X)}$:

$$\min_{\beta^\top\beta=I_d,\,\{a_j,b_j\}} \sum_{j=1}^{n}\sum_{i=1}^{n}\Big(Y_i - \{a_j + b_j^\top\beta^\top(X_i - X_j)\}\Big)^2 w_{ij}$$

where $w_{ij} = K_h(X_i - X_j)/\sum_{\ell=1}^{n} K_h(X_i - X_\ell)$.

Our CSDR Approach

CSDR Algorithm (Hsiao et al. 2026):

1. Use $O = (Y, X, C)$ to estimate $\widetilde{Y}$.
2. Estimate $\mathcal{S}_{\mathbb{E}(Y^x)}$ by running MAVE on $(\widetilde{Y}, X)$.
3. Estimate $\mu(x)$ by replacing X with $Z := \widehat{\beta}^\top X$.

$$\widehat{\mu}_{DR,\widehat{\beta}}(z) = \frac{1}{n}\sum_{i=1}^{n}\left\{\frac{K_h(\widehat{Z}_i - z)}{\widehat{\pi}_{\widehat{\beta}}(\widehat{Z}_i \mid C_i)}\left[Y_i - \widehat{m}_{\widehat{\beta}}(\widehat{Z}_i, C_i)\right] + \widehat{m}_{\widehat{\beta}}(z, C_i)\right\},$$

4. Construct 95% confidence intervals for $\widehat{\mu}_{DR,\widehat{\beta}}(z)$ based on empirical variance of the influence function.

Alternative approach

The CSDR approach improves on the MSM approach by avoiding numerical optimization on β-dependent nuisances.

However, estimation of $\widetilde{Y}$ still involves nuisance estimation of $m(x, c)$ and $\pi(x \mid c)$.

We propose another alternative that can estimate $\mathcal{S}_{\mathbb{E}(Y^x)}$ without $m(x, c)$ and $\pi(x \mid c)$.

Partial dimension reduction on $\mathbb{E}(Y \mid X, C)$

For this approach, we make an additional assumption that

$$\mathbb{E}(Y^x \mid C = c) = g(\beta^\top x, c).$$

One can show that any β satisfying the above also satisfies our original central assumption:

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It follows that the partial central mean subspace of Y given (X, C) is equal to $\mathcal{S}_{\mathbb{E}(Y^x)}$.

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Solution: Run partial dimension reduction (Li et al. 2010) directly on $\mathbb{E}(Y \mid X, C)$.

Theoretical results: subspace estimation

Let $\delta_n = \sqrt{\log n/(nh^d)}$ and $R_n = \|\hat{\mu}(X) - \mu(X)\|_\infty$ (the error in estimating $\widetilde{Y}$).

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1. **Subspace estimation**: We establish convergence rates for estimating $\mathcal{S}_{\mathbb{E}(Y^x)}$:

- Method 1: $\|\widehat{\beta} - \beta\| = O_p(h^3 + h\delta_n + h^{-1}R_n)$ (dependent on $\widetilde{Y}$ estimation)
- Method 2: $\|\widehat{\beta} - \beta\| = O_p(h^3 + h\delta_n + h^{-1}\delta_n^2)$ (nuisance estimation-free)

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2. **Structural dimension selection**: A cross-validation procedure consistently estimates the dimension of $\mathcal{S}_{\mathbb{E}(Y^x)}$.

$$\mathbb{P}(\hat{d} = d_0) \longrightarrow 1.$$

Theoretical results: asymptotic equivalence

One may wonder how estimating $\mu(x) = \mathbb{E}(Y^x)$ using $\widehat{\beta}_n$ compares to the true β.

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Asymptotic equivalence: We show that error in estimating β can be *asymptotically negligible* for estimating $\mu(x)$ (Forzani et al. 2024):

$$\sqrt{nh_n^d}\{\hat{\mu}_{DR}(x;\hat{\beta}_n) - \hat{\mu}_{DR}(x;\beta)\} = o_p(1)$$

Insight: Estimation of β can achieve $O_p(n^{-1/2})$ convergence, faster than the $O_p\{(nh^d)^{-1/2}\}$ rate of $\hat{\mu}_{DR}(x)$.

Theoretical results: subspace importance scores

Problem: β is not uniquely identified, so direct interpretation of loadings is tricky.

Solution: Measure each exposure's contribution using the projection matrix, which is unique.

$$\lambda_j = e_j^\top \beta(\beta^\top \beta)^{-1}\beta^\top e_j = \left[\beta(\beta^\top \beta)^{-1}\beta^\top\right]_{jj}$$

- Basis invariant: unaffected by rotations in β.
- Bounded: $0 \leq \lambda_j \leq 1$
- Decomposition: $\sum_{j=1}^{p} \lambda_j = d$

Summary of our work

Existing approaches either

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Our approach builds on and combines the strengths of these previous approaches

- Flexibly estimates the nonparametric dose-response in a lower dimension
- Returns a causally and scientifically interpretable β
- Final dose-response estimation is asymptotically robust to estimation error of $\widehat{\beta}$

Simulations and Real Data

CSDR recovers the true $\mathcal{S}_{\mathbb{E}(Y^x)}$

Table 1: Frobenius norm error between estimated and true $\mathcal{S}_{\mathbb{E}(Y^x)}$ with estimated $\hat{d}$. Mean (SD).

Method	$n = 500$	$n = 1000$	$n = 5000$
CSDR	0.45 (0.39)	0.22 (0.29)	0.03 (0.04)
pCCA	1.20 (0.18)	1.10 (0.20)	0.74 (0.20)
MAVE	1.29 (0.17)	1.17 (0.10)	1.07 (0.02)
PCA	2.00 (0.00)	2.00 (0.00)	2.00 (0.00)

CSDR recovers the true structural dimension

Table 2: Proportion selecting each dimension ($d_0 = 2$).

n	Method	$d = 1$	$d = 2$	$d = 3$	$d = 4$
500	**CSDR**	0.03	**0.88**	0.07	0.01
	MAVE	0.00	**0.11**	0.75	0.14
5000	**CSDR**	0.00	**1.00**	0.00	0.00
	MAVE	0.00	**0.00**	1.00	0.00

CSDR improves inference of the causal dose-response

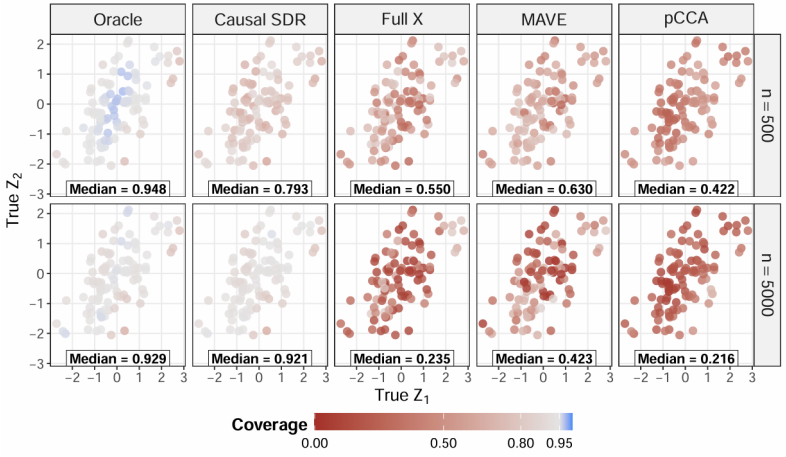


Figure 1: Pointwise 95% coverage for the exposure-response curve ($p = 10$, $d = 2$) at 100 test points; methods differ only in β estimation.

PFAS real data application

We use the **Atlanta African American Maternal-Child Cohort** ($n = 305$) to study the effect of a PFAS mixture ($p = 4$) on infant birthweight (Chang et al. 2022). Covariates include maternal age, BMI, education, tobacco use, marijuana use, and infant sex.

ESTIMATED REDUCTION ($\widehat{d} = 1$)

$$\hat{Z} = 0.77\,\text{PFOS} + 0.14\,\text{PFOA} + 0.26\,\text{PFNA} - 0.56\,\text{PFHxS}$$

SUBSPACE IMPORTANCE SCORE $\widehat{\lambda}_j$

PFOS	0.60
PFOA	0.02
PFNA	0.07
PFHxS	0.31

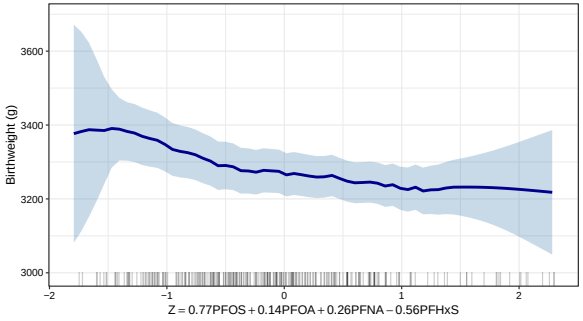
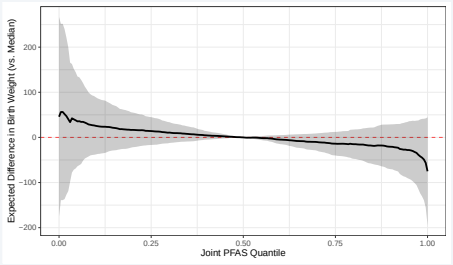


Figure 2: PFAS exposure-response curve from ATL-AA

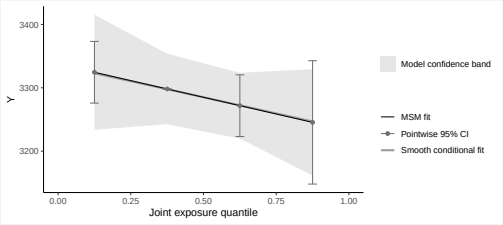
BKMR and qgcomp comparison analysis

BKMR



All four PFAS share a quantile versus the median.

qgcomp



All four PFAS shift together across exposure quartiles.

Takeaway: Different estimands, but in general, increasing PFAS leads to decreasing birthweight.

Some thoughts on mixtures

My opinion

I believe the biggest issues in causal inference for environmental mixtures are positivity violations.

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Questions?



CSDR preprint

arxiv.org/abs/2606.14840



csdr R package

github.com/tXiao95/csdr

Appendix

Marginal structural model approach

Using the theory of marginal structural models (Robins 2000), Nabi et al. (2022) proposed estimating $\mathcal{S}_{\mathbb{E}(Y^x)}$ by solving the following efficient score:

$$
\begin{aligned}
0 = \frac{1}{n}\sum_{i=1}^{n}\Bigg[& \frac{\pi^*(X_i)}{\pi(X_i \mid C_i)}\Big\{Y_i - \mathbb{E}(Y \mid X_i, C_i)\Big\} \\
& \times \Big\{\alpha(X_i) - \mathbb{E}_q[\alpha(X) \mid \beta^\top X = \beta^\top X_i]\Big\} \\
& + \int_{\mathcal{X}} \Big\{\mathbb{E}(Y \mid X = x, C_i) - \mathbb{E}_q[Y \mid \beta^\top X = \beta^\top x]\Big\} \\
& \times \Big\{\alpha(x) - \mathbb{E}_q[\alpha(X) \mid \beta^\top X = \beta^\top x]\Big\}\pi^*(x)\, dx\Bigg].
\end{aligned}
$$

Marginal structural model approach difficulties

While the method is elegant, practical implementation is difficult.

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Nuisance functions to be estimated:

1. $m(x, c) := \mathbb{E}(Y \mid X, C)$
2. $\pi(x \mid c) := f_{X|C}(x \mid c)$
3. $\mathbb{E}_q(Y \mid \beta^\top X = z)$
4. $\mathbb{E}_q[\alpha(X) \mid \beta^\top X = z]$

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4. $\mathbb{E}_q[\alpha(X) \mid \beta^\top X = z]$

Problem: Optimization is challenging when the parameter is matrix-valued and nuisance functions must be re-estimated at each iteration of β.

Setup outcome, exposures, and confounders

R code

```r
library(csdr)
data(csdr_example2)

Y <- csdr_example2$Y
A <- csdr_example2$A
C <- csdr_example2$C

head(Y, 5)
head(A[, c(1:3, 18:20)], 5)
head(C, 5)
```

Console

```
> head(Y, 5)
     obs1      obs2      obs3      obs4      obs5
  0.3805    0.1838   -2.0486    0.4977    5.4798

> head(A[, c(1:3, 18:20)], 5)
         A1     A2     A3    A18    A19    A20
obs1 -1.066  1.573  0.953  0.033 -0.480  0.500
obs2 -1.028  0.325 -0.364 -0.967 -1.543 -0.572
obs3 -0.800 -0.103 -0.433 -0.011  0.505  0.623
obs4  2.492  0.582 -1.134 -1.473 -1.138  1.551
obs5  1.733 -0.114  0.615  1.083  0.350  0.133

> head(C, 5)
         C1    C2     C3     C4     C5
obs1 -0.935 1.312 -0.294 -1.979  1.150
obs2 -0.694 0.531  0.367 -2.259 -1.293
obs3 -0.548 0.712 -0.088 -0.407  1.828
obs4  0.766 1.050 -0.498 -0.593 -0.604
obs5  1.852 -0.554  0.849  0.643 -0.312
```

Fit the causal subspace

R code

```r
sl_lib <- c(
  "SL.glm", "SL.earth", "SL.ranger"
)

learners <- csdr_learners(
  sl_library = sl_lib
)

set.seed(42)
fit <- csdr(
  Y = Y, A = A, C = C,
  L = 5, seed = 42,
  learners = learners,
  max_dim = 5,
  keep_nuisance = TRUE
)

summary(fit)
```

Console

```
> summary(fit)
Summary of causal sufficient dimension reduction fit

 variant d_hat  n p mave_method target_exposure
      DR     2 100 20   meanMAVE               A

Learners:
 outcome: SuperLearner regression
   family: gaussian
   SL.library: SL.glm, SL.earth, SL.ranger
 gps: MVN GPS with linear conditional mean
   method_gps: linear; density_floor: 1e-16

Target construction:
Effective folds: 5
PO marginalization: crossfit
```

Inspect the fitted subspace and scores

R code

```r
B_hat <- coef(fit)
Z_hat <- scores(fit)

mave <- mave_fits(fit)
dim_selection <- mave$dimension_selection

B_hat[c(1, 2, 20), , drop = FALSE]
sis(fit)
head(Z_hat, 2)
dim_selection
```

Console

```
> B_hat[c(1, 2, 20), , drop = FALSE]
       dir1    dir2
A1  -0.3401  0.5777
A2  -0.3524  0.4195
A20 -0.0521 0.0784

> sis(fit)
    A1    A2    A3    A4    A5    A6    A7    A8    A9   A10
 0.464 0.312 0.533 0.390 0.000 0.004 0.002 0.020 0.001 0.021
   A11   A12   A13   A14   A15   A16   A17   A18   A19   A20
 0.068 0.059 0.028 0.020 0.006 0.005 0.015 0.036 0.006 0.009

> head(Z_hat, 2)
       score1  score2
obs1  -0.1862  0.4515
obs2   1.5224 -0.3484

> dim_selection
Dimensions      0    1    2    3    4    5
CV-value     1.58 1.13 0.75 0.77 0.92 0.97
Selected dimension: 2
```

Estimate the response in the reduced exposure

R code

```r
z_grid <- matrix(
  rep(apply(Z_hat, 2, median), each = 50),
  nrow = 50,
  dimnames = list(NULL, colnames(Z_hat))
)
z_grid[, 1] <- seq(
  quantile(Z_hat[, 1], 0.05),
  quantile(Z_hat[, 1], 0.95),
  length.out = 50
)

ers <- estimate_ers(
  Y = Y, A = Z_hat, C = C,
  a_eval = z_grid, estimator = "DR",
  L = 5, seed = 42,
  verbose = FALSE
)

head(ers$results)
```

Console

```
> head(ers$results)
  score1 score2 estimate     se ci_lower ci_upper effective_n
1 -1.990 0.0015    2.676 0.5599    1.579    3.774       4.502
2 -1.912 0.0015    2.693 0.4762    1.760    3.627       4.559
3 -1.834 0.0015    2.703 0.3920    1.935    3.472       4.671
4 -1.756 0.0015    2.703 0.3136    2.088    3.317       4.851
5 -1.678 0.0015    2.687 0.2493    2.198    3.176       5.113
6 -1.601 0.0015    2.652 0.2100    2.241    3.064       5.476
    min_gps h_used_1 h_used_2
1 4.049e-05    0.578    0.596
2 4.049e-05    0.578    0.596
3 4.049e-05    0.578    0.596
4 4.049e-05    0.578    0.596
5 4.049e-05    0.578    0.596
6 4.049e-05    0.578    0.596
```

Dimension Selection

NW leave-one-out estimator

$$\hat{a}_{d0,j} = \frac{\sum_{i \neq j} K_{h_d}^{(i,j)} \tilde{Y}_i}{\sum_{i \neq j} K_{h_d}^{(i,j)}}, K_{h_d}^{(i,j)} = K_{h_d}\{\hat{\beta}_1^\top (\tilde{X}_i - \tilde{X}_j), \ldots, \hat{\beta}_d^\top (\tilde{X}_i - \tilde{X}_j)\}$$

Cross-validated error

$$\mathsf{CV}_n(d) = \frac{1}{n} \sum_{j=1}^{n} \left(\tilde{Y}_j - \hat{a}_{d0,j}\right)^2, \qquad \mathsf{CV}_n(0) = \frac{1}{n} \sum_{i=1}^{n} (\tilde{Y}_i - \bar{Y})^2$$

Estimated structural dimension

$$\hat{d} = \arg \min_{0 \leq d \leq p} \mathsf{CV}_n(d)$$